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RESEARCH ARTICLE

A Hybrid Recommendation System: A Review

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ABSTRACT Over the years, recommendation engines (REs) have steadily increased in popularity, providing significant advantages by matching available items to users' specific interests. As more users, products, and rating information are introduced into the system, the relationship between users and recommended products changes, leading to a phenomenon known as concept drift (CD), which degrades the system's accuracy. Deep learning (DL), a subset of machine learning methodologies, involves learning over multiple layers of information processing phases and aids in addressing the CD problem. Additionally, federated recommender systems (FedRecSys) provide privacy for user data within a decentralised framework. This paper proposes a novel approach that hybridises DL methods with federated learning to detect and adapt to the concept drift issue in E-commerce-based REs. It also focuses on the benefits of collaborative RE and provides a systematic literature review of hybrid federated DL models to solve the challenges posed by concept drift for accurate recommendations. The results can be evaluated using performance measures such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Squared Log Error (MSLE), among others.

INDEX TERMS Recommendation engines, machine learning, big data, content-based filtering, collaborative filtering, deep learning, concept drift, concept diagrams.

I. INTRODUCTION

A. BIG DATA

Big data is a vast informational repository that businesses utilise to optimise product selection, pricing, and advertising to increase revenues. Big data analytics—characterised by the six Vs (volume, variety, velocity, veracity, validity, and volatility) [1]—have emerged as a powerful tool for processing large amounts of data and enhancing decision-making. Volume, the amount of information needed and analysed to produce the desired outcome, is the primary attribute. It gathers information from cameras, scanners, mobile devices, digitisers, video, email, social media, and the internet. Variety refers to the different types of data analysed, processed, and stored, such as text, images, videos, sounds, geometrics, and combinations of these types [1]. Veracity describes whether data is sufficiently accurate for use in big data [1]. Online marketplaces such as Amazon, eBay, and Etsy have expanded significantly in recent years. Due to the wide variety of products offered, these platforms primarily rely on Recommendation Engines (REs) to guide consumers towards products that match their preferences and interests. These recommendation methods typically maximise relevance by estimating an item's likelihood of being purchased or clicked [1].

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B. RECOMMENDATION ENGINES (RE)

Presently, REs are widely used in e-commerce for content delivery and product recommendations. Previous research often represents item content using independent features, thus ignoring the hidden connections between content aspects. Only a select few sets of features may be responsible for the user appeal of an item [21]. In-depth research on recommendation models has been extensively covered in the literature, primarily focusing on two features: 1) user interest, derived from user ratings, purchase history, click-through activities, or sharing habits, and 2) item content, which includes item attributes [15]. Three basic recommendation paradigms have emerged regarding the use of various features: collaborative filtering (CF), content-based filtering (CBF), and methods

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that combine the two, i.e., the hybrid filtering technique, as shown in FIGURE 1 [2], [15].

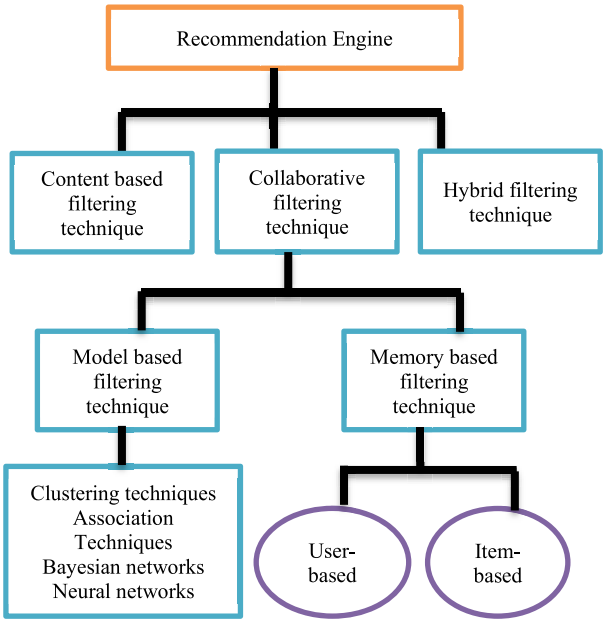


FIGURE 1. Filtering-based recommender system hierarchy.

C. RESEARCH BACKGROUND

FIGURE 2 presents the various categories of REs as memory-based (heuristic and non-parametric) and model-based (algorithmic and parametric) methods.

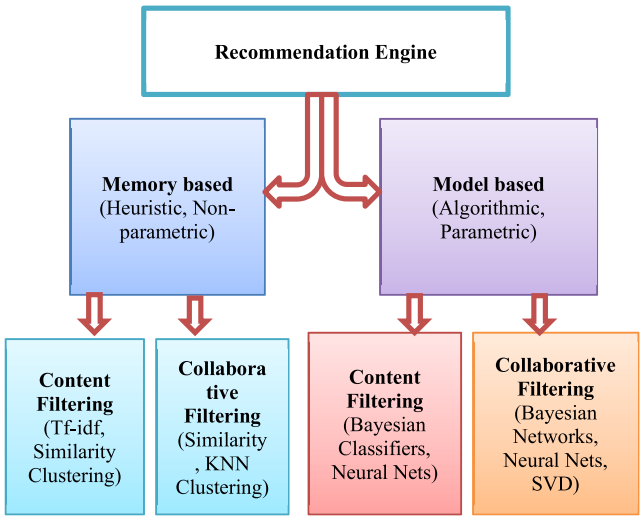


FIGURE 2. Categories of Recommendation Engines (REs).

When the RE is more familiar with the characteristics of users and objects, it produces better results [4]. For a specific period, evaluating recommendations based on impact, interaction, and conversion are well-known metrics for assessing the efficacy of systems, particularly in e-commerce platforms [4].

The outcomes of model-based procedures are simpler to comprehend and interpret. The learning of parametric models from data happens quickly, requiring less training data while still performing well even if the data fit is not ideal [4]. Conversely, memory-based techniques require more training data to estimate the mapping function. These techniques often have many more parameters, leading to longer training times. The training data are more likely to be overfit, making it more challenging to explain and provide examples of specific predictions [4].

REs face numerous issues, such as concept drift (CD), cold start, data sparsity, the grey sheep problem, and more. The most prominent issue that needs to be addressed in REs is CD, which causes data drifts in the dynamic, streaming, non-stationary, and large datasets typical of e-commerce REs. These drifts affect the accuracy, efficiency, and performance of the overall system, resulting in poor customer satisfaction and trust.

To address this major issue in REs, we propose a novel approach that hybridises the collaborative RE (CRE)—the best recommendation algorithm according to the literature review—and federated learning, an emerging dimension of deep learning. This new model, named Federated Recommender Systems (FedRecSys), detects and adapts to concept drift through the federated learning mechanism. Federated learning provides security and privacy for users' private data within a decentralised machine learning framework in e-commerce REs.

Therefore, this novel approach of FedRecSys offers accurate recommendations, ultimately gaining customer/user satisfaction and trust.

The rest of the paper is organised as follows: Section II focuses on the systematic literature review in three parts: (a) recommendation systems as a model, (b) machine learning and recommendation systems for optimization, and (c) information retrieval and recommendation systems for effective information retrieval.

Section III presents various comparative studies, including: (a) Concept drift detection in REs using federated learning, (b) top-rated websites, (c) types of REs, (d) survey methodology (PRISMA), (e) challenges in REs, (f) challenges in CREs, (g) concept diagram (gap analysis), and (h) systematic literature review on hybrid deep learning models and concept drift issues.

Section IV discusses Federated Learning in terms of FedRecSys and provides a detailed literature review of FedRecSys. Section V focuses on research methodology. Section VI presents the literature review of RE datasets. Section VII elaborates on the dataset design, such as the Amazon reviews/ratings dataset. Section VIII presents the overall outcomes of the work proposed in the manuscript through the conclusion, followed by the references.

II. RESEARCH CONTRIBUTION-PART I

Several research studies highlight the importance of REs, their various real-life applications, datasets, and

TABLE 1. Systematic literature review of RS as business model, ML & RS and IR & RS.

Authors	Year	Contribution	Outcomes
Recommendation System as Business Model: Representation of Recommender System as a Business Model			
Alamdari, et al. [4]	2020	Comprehensive and systematic literature review (SLR) regarding the research conducted in the field of e-commerce recommender systems.	Content-based filtering (CBF), collaborative filtering (CF), demographic-based filtering (DBF), hybrid filtering, and knowledge-based filtering (KBF) are the five categories of RSs algorithms that have been highlighted in this work.
Lindgren, et al. [5]	2018	this study addressed when a business model can be classified as a sensing business model (SBM). Additionally, it discussed when a business can be classified as one involved in sensing business model innovation (SBMI).	The goal of the research was to contribute to the creation of conceptual framework models and typologies for classifying disruptive, radical, and gradual SBMIs, as well as SBMs.
Singh, et al. [6]	2020	Presented a thorough analysis on the RS and covered various recommendation approaches, related problems, and information retrieval techniques.	The study projected RS's future, which could pave the way for new areas of study in this field.
Lindgren, et al. [7]	2011	Presented an innovative and futuristic perspective on the creation of business models that are environmentally friendly.	Clarified the procedures for developing new business models for a future green society.
Machine Learning and Recommendation System for Optimization			
Sardianos, et al. [8]	2019	It suggested that CF algorithms, which run concurrently on several graph partitions, might perform better if a subset of users and their ratings were duplicated between partitions.	Improvisation in the collaborative filtering algorithms.
Tran, et al. [9]	2020	Explored Möbius gyrovector spaces, where the formalism of the spaces might be used to generalise the most prevalent Euclidean vector operations, with the goal of focusing on the research of the relationship between metric learning in hyperbolic space and collaborative filtering.	Demonstrated that compared to other recommendation systems, the collaborative filtering method performs better.
Louca, et al. [10]	2019	Proposed a cutting-edge revenue model that simultaneously maximises profit and purchase likelihood.	Demonstrated that the recommendations generated by the suggested methodology can boost platform revenue while maintaining high user relevancy.
Information Retrieval and Recommendation System for Effective Information Retrieval			
Raza, et al. [11]	2019	Highlighted the research aimed at helping the CARS research community understand how context can be incorporated into traditional recommendation systems.	The main objective was to reduce the dimensionality of various contexts by integrating context into the recommendation algorithms.
George [12]	2019	Conventional recommender systems that have been presented and have been successful in e-commerce and other domains cannot be used in the e-learning sector. This study demonstrated an ontology-based recommender system, despite the fact that other methods are available to create suggestions on e-learning.	The ontology-based recommender system enables for the inclusion of more learner and learning object specifics than the fundamental models of recommender systems.

issues/challenges. In this study, the research work is presented through an in-depth systematic literature review (SLR).

The SLR is organised into three sections, as shown in TABLE 1:

- -Recommendation system (RS) as a business model
- -Machine learning and recommendation system for optimization
- -Information retrieval and recommendation system for effective information retrieval

III. RESEARCH CONTRIBUTION-PART II-METASURVEY

In this section, various comparative analyses and systematic literature reviews are presented as follows:

i. Comparison of related studies in machine learning (ML), recommendation systems (RS), deep learning (DL), concept drift (CD), and federated learning (FL).
ii. Comparison of top-rated websites.
iii. Types of REs.

TABLE 2. Systematic review of concept drift detection in RSs using federated learning (√: Yes, X: No).

Authors	Year	ML	RS	DLRS (Hybrid)	CD	DL + CD (Hybrid)	FL
Sardianos et al [8]	2019	√	√	X	X	X	X
Tran et al [9]	2019	√	√	X	X	X	X
Louca et al [10]	2019	√	√	X	X	X	X
Manikandan, et al [14]	2019	√	√	X	X	X	X
Sahoo et al [15]	2019	√	√	X	X	X	X
Zhang et al [16]	2019	√	√	X	X	X	X
Jameel, et.al [22]	2020	X	X	X	X	√	X
Disabato et. al [23]	2021	X	X	X	X	√	X
Yang et. al [24]	2021	X	X	X	X	√	X
Hashmani, et.al [25]	2020	X	X	X	X	√	X
Lin, et.al [26]	2019	X	X	X	X	√	X
Wares, et.al [27]	2019	X	X	X	X	√	X
Rabiu et. al [31]	2020	X	X	X	√	X	X
Cakir [51]	2020	X	X	√	X	X	X
Dong et al [52]	2017	X	X	√	X	X	X
Sadegh et al [53]	2021	X	X	√	X	X	X
Huang et al [54]	2019	X	X	√	X	X	X
Çano [55]	2017	X	X	√	X	X	X
Chen et al [56]	2018	X	√	X	X	X	X
Hors-Fraile et al [57]	2018	X	√	X	X	X	X
Nielsen et al [58]	2018	X	√	X	X	X	X
Wanas et. al [59]	2019	X	X	X	√	X	X
Mothukuri et. al [60]	2021	X	X	X	X	X	√
Li et. al [61]	2020	X	X	X	X	X	√
Zhang et. Al [62]	2021	X	X	X	X	X	√
Cao et.al [63]	2022	X	X	X	X	X	√
Amalraj et.al [64]	2022	X	X	X	X	X	√
Jere et.al [65]	2021	X	X	X	X	X	√
Li et.al [66]	2020	X	X	X	X	X	√
Asad et.al [67]	2020	X	X	X	X	X	√
Our Study	**2022**	√	√	√	√	√	√

iv. Survey methodology (PRISMA).
v. Literature review on challenges in RE
vi. Literature review on challenges in CRE.
vii. Concept diagram (gap analysis).
viii. SLR on hybrid deep learning models and concept drift issues.

The above meta-survey provides a detailed study of the types of recommendation algorithms used in well-known e-commerce websites. It also facilitates an in-depth analysis of various issues in REs, such as accuracy and scalability.

Additionally, the meta-survey focuses on the hybridization of deep learning models and concept drift issues, along with the related outcomes.

i) TABLE 2 provides a systematic review with an in-depth analysis and future recommendations for concept drift detection in recommendation systems using federated learning.
ii) The top-rated websites can be compared with respect to:
a) Techniques for recommendation system or engine

TABLE 3. Comparison of top-rated websites for recommendation techniques.

Sr. No.	Website	Algorithm Used	Recommendation Performance
1	Amazon	Item-Item Collaborative Filtering	Here, the items, rather than the users, form the basis of the recommendation system's operation. The cosine similarity function is used to generate recommendations by sorting the user's list of visited items into an item-to-item matrix. Amazon recommends similar items of different shapes, sizes, and brands to users based on the types of items they have searched for or viewed earlier [2,13].
2	Facebook	Collaborative Filtering (Matrix Factorization)	Facebook recommends friends based on the list of mutual connections or recommendations based on compatible people [2,13].
3	Flipkart	Hybrid filtering (Content-based (CB)+Collaborative Filtering (CF))	A Content-Based algorithm is utilised for item attributes and photos in the catalog, while a Collaborative Filtering algorithm tracks a user's search history, including pages visited, items added to a wish list or shopping cart, and more, to identify items that are frequently searched collaboratively for a particular product [2,13].
4	Netflix	Machine Learning technique	Netflix recommends video content based on the user's watch records, including the time duration spent watching a video and the device used to perform the action [2,13].
5	YouTube	Deep Neural Networks	Here, videos are selected for recommendation from hundreds of options after retrieving the implied data, such as the user's watch history [2,13].

b) Items or products and facilities recommended by the websites [2], [13], as given in TABLE 3:

iii) Types of REs:

Nowadays, people prefer to buy items online rather than in stores. Consumers once relied on recommendations from relatives or friends, but with the proliferation of digital options, it is essential to assure them that a product is worthwhile and enjoyable. Recommender systems were created to instil confidence in purchasing decisions [18]. These systems are among the most popular and widely used applications of machine learning in the commercial world. Recommendation systems help businesses boost revenue by enabling customers to find the best products for their needs [3], [18].

The different types of REs are described in detail below.

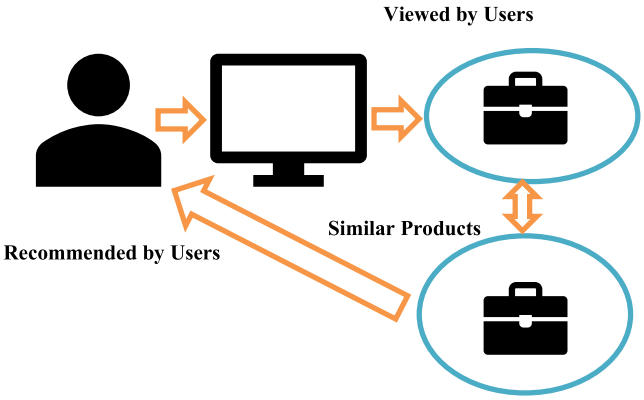


FIGURE 3. Content-Based Filtering (CB).

A. CONTENT-BASED FILTERING RE

FIGURE 3 focuses on the content-based filtering RE.

This recommendation method centres on the products themselves, suggesting additional items with similar characteristics. Content-based filtering relies on the product's attributes and does not depend on other users' interactions to generate suggestions [3], [6].

B. DEMOGRAPHIC-BASED FILTERING RE

FIGURE 4 focuses on the demographic-based filtering RE.

This recommendation system categorises users into groups based on their demographics. It requires market research data. Its key advantage is that it does not depend on a history of user ratings [3], [6].

C. UTILITY-BASED FILTERING RE

FIGURE 5 focuses on the utility-based filtering RE.

This technology provides suggestions based on a calculation of the value to each user, contingent on the industry's ability to define a user-specific utility function. The key benefit of this system is its capability to recommend product qualities such as availability and vendor trustworthiness [3], [6].

D. KNOWLEDGE-BASED FILTERING RE

FIGURE 6 focuses on the knowledge-based filtering RE.

This system generates recommendations based on information about the interests and needs of each user.

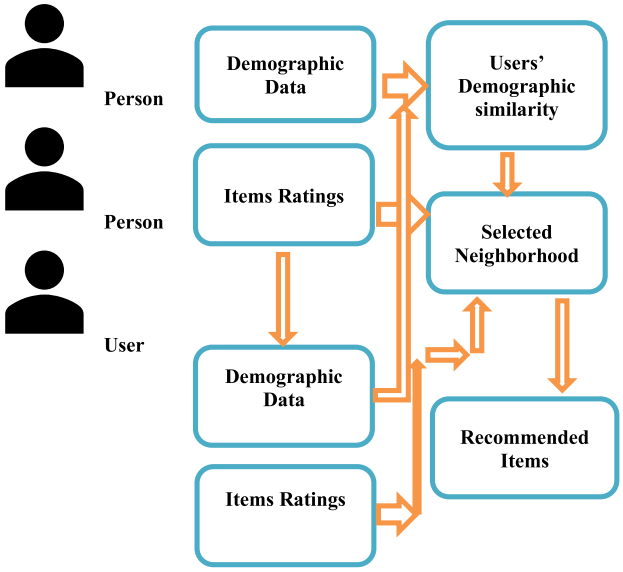


FIGURE 4. Demographic-Based Filtering (DB).

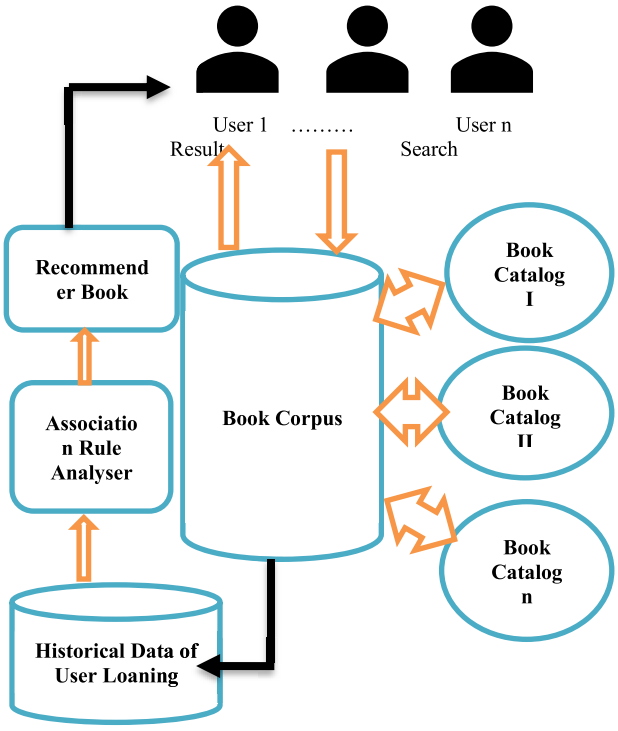


FIGURE 5. Utility-based filtering (UB).

Understanding these functions helps connect a customer's requirements with suitable products [3], [6].

E. COLLABORATIVE FILTERING RE

FIGURE 7 represents CRE.

The collaborative filtering approach predicts unknown outcomes by generating a user-item matrix of choices or preferences for items [20]. This matrix is compared with the user's preferences and interests to assess the similarity of profiles. User groups form the neighbourhood, and users who have not previously rated certain items receive

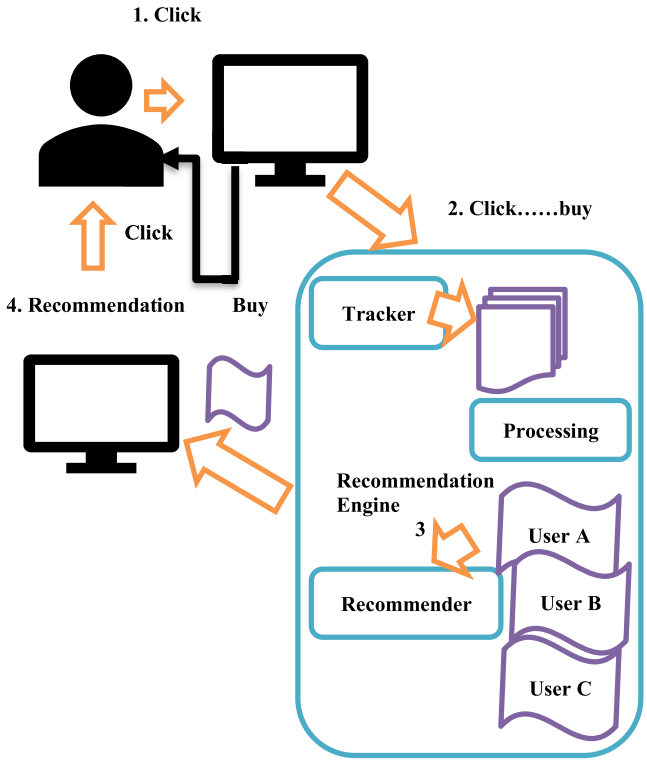


FIGURE 6. Knowledge-Based Filtering (KB).

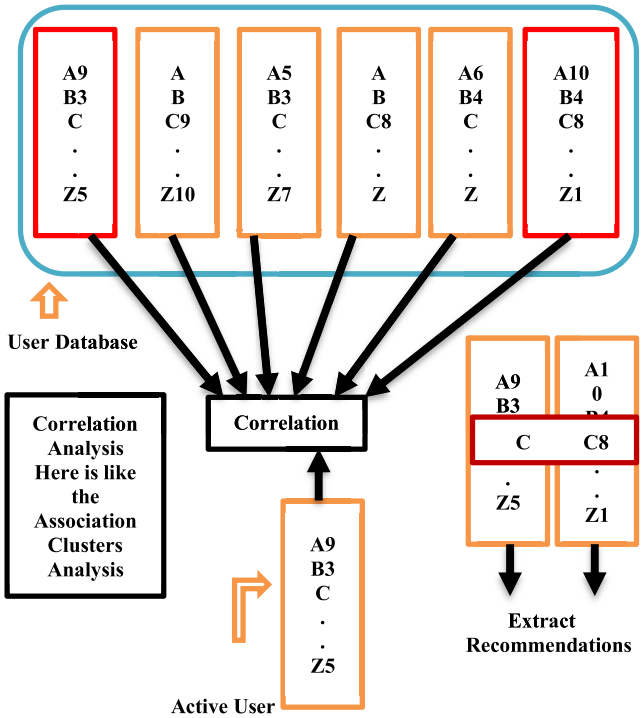


FIGURE 7. Collaborative Recommendation Engine (CRE).

recommendations for those items based on positive ratings from others in their group. In a recommendation system, collaborative filtering can be used for both prediction and suggestion [3], [20].

F. HYBRID FILTERING RE

FIGURE 8 focuses on the hybrid Filtering approach.

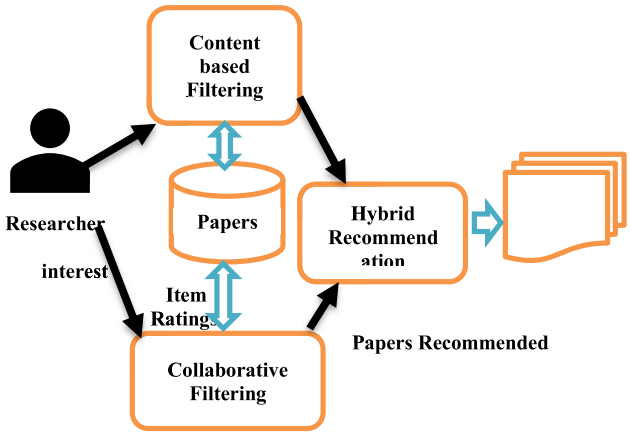


FIGURE 8. Hybrid Filtering (HB).

- This recommendation system can integrate any combination of two or more recommendation systems [6]. The hybrid recommendation system employs three fundamental strategies: Post-fusion, Middle-fusion, and Former-fusion [12], [29].
- In the post-fusion approach, the outputs of multiple recommendation algorithms are combined using a voting mechanism, linear combination, or confidence level to generate the final recommendation results [12].
- In the Middle-fusion method, one recommendation algorithm is combined with another [12].
- In the Former-fusion method, various recommendation algorithms are integrated into a single recommendation model, which is then trained on a diverse set of samples [12], [29].

Monolithic hybrid recommendation, parallel hybrid recommendation, and pipeline hybrid recommendation are the three types of hybrid recommendation systems. A monolithic hybrid recommendation integrates multiple recommendation systems into a single algorithm. The other two types require at least two different recommendation systems, which must be combined [38].

Traditional hybrid recommender systems require significant manual effort for feature engineering and often struggle to extract deep-dimensional information for users and items, which limits the effectiveness and scalability of hybrid recommendations. The nonlinear architecture of deep neural networks, however, enables the automatic retrieval of latent or hidden properties from large datasets, thereby saving time and effort [12], [29].

iv) Survey Methodology

The preferred reporting items for systematic reviews and meta-analyses (PRISMA) [49] guidelines were used as the foundation for the current systematic literature review. Kitchenham's [50] standard procedures were followed, and FIGURE 9 shows the study screening and selection procedure. TABLE 4 illustrates the SLR method.

TABLE 4. SLR method based on PRISMA protocol.

Title	Description
Abstract	Gives a summary of the study, including its context, methods, and finding
Introduction	Gives a summary of the current state of knowledge in relation to Big Data, REs, and different types of REs.
Methodology	Research question Information source and database Search strategy and key terms Eligibility criteria Quality assessment Data extraction
Conclusion	Provide conclude outcomes of the entire research study

Research Motivations:

Concept drift detection in REs using FLis a rapidly evolving methodology. Consequently, an in-depth examination of prior studies is necessary. Our research aims to identify the most effective approaches for detecting concept drift issues and to explore current methods for addressing them using advanced DL techniques. This study will provide readers with insights into various concept drift challenges in REs, especially those that are emerging in the field. Some investigations have addressed concept drift using traditional methods and DL techniques. Our work focuses on DL and federated learning approaches for concept drift, emphasizing reliable and precise recommendations by REs. Additionally, we concentrate on studies related to resolving concept drift with FL techniques, as depicted in FIGURE 9.

V) Challenges in RE:

RE is advantageous for recommendations, although it faces several challenges. Some of these issues are discussed in TABLE 5.

TABLE 5. Challenges in recommendation systems.

References	Data Collection	Data Sparsity	Scalability	User Trust	Grey Sheep Problem	Synonymy	Privacy & Security
[14]	√						
[15]		√					√
[16]		√					
[17]	√						
[18]		√	√				√
[19]	√	√	√	√	√	√	√

vi) Challenges in CRE:

TABLE 6 presents challenges in CREs, with highlighted text indicating the targeted focus areas for the proposed research work.

vii) Concept Diagram (Gap Analysis)

FIGURE 10 presents a pictorial representation of the outcome of the systematic literature survey as a gap analysis.

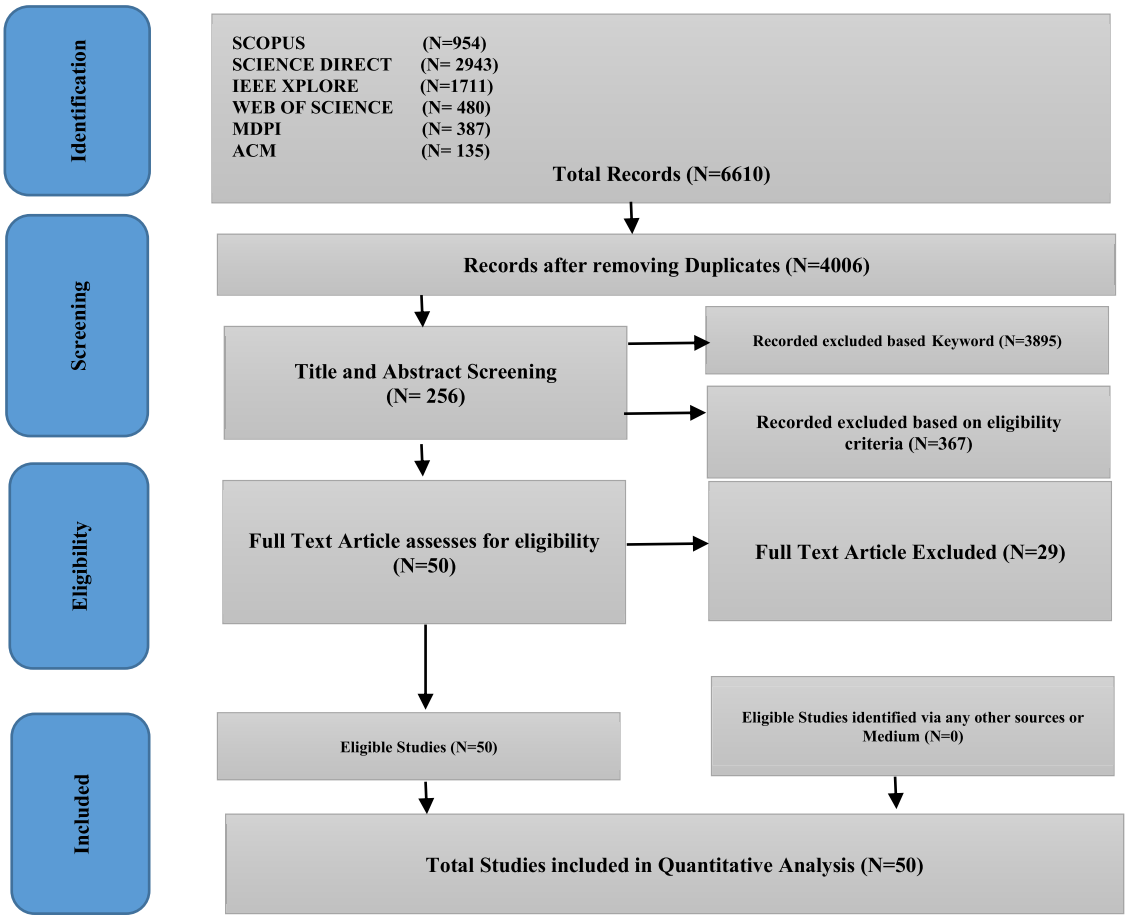


FIGURE 9. Diagram illustrates the screening and choice of research using PRISMA.

TABLE 6. Challenges in collaborative recommendation systems.

References	Data Collection	Data Sparsity	Scalability	User Trust	Synonymy	Privacy & Security
[20]	√		√	√		
[4]	√	√	√	√		√
[6]		√		√	√	
[21]				√		

Machine Learning Gaps:

Following are the limitations of machine learning (ML) algorithms:

a) Optimization was required for massive datasets.
b) To target real time data.
c) Issues of data sparsity and diversified learner's behaviour.

Several points were discussed in the previous study to address these issues in the ML framework:

(i) Validating ongoing cross-domain application work and developing a plan to integrate it with other techniques using actual data [8].
(ii) Simulating different learner behaviours to better adjust recommendations to various learning methods [8].
(iii) Applying DL algorithms to big data applications in the future [8].

Given the gaps and challenges faced in ML when working with e-commerce platforms and dynamic real-time data, REs plays a crucial role by addressing these challenges through various outcomes, such as user-item preferences, association/correlation matrices, past records, parallel execution, and exploration of E-Commerce platforms. These efforts result in optimised, ranked, and relevant recommendations for users. However, despite the various types of REs and hybrid REs shown in FIGURE 10, issues such as data sparsity, cold start, and concept drift remain.

DL models (DLMs), such as convolutional neural networks [(CNNs), recurrent neural networks (RNNs),], autoencoders (AEs], deep belief network (DBN), neural collaborative filtering (NCF), and probabilistic matrix factorization (PMF), address issues related to massive data, latent (hidden) features, and sparse data. Nevertheless, hybrid DLMs still face the challenge of concept drift, which involves variations in data distribution over time. Recent research has focused on addressing the issue of concept drift with DL and REs.

viii Literature Review (Hybrid Deep Learning Models and Concept Drift Issue)

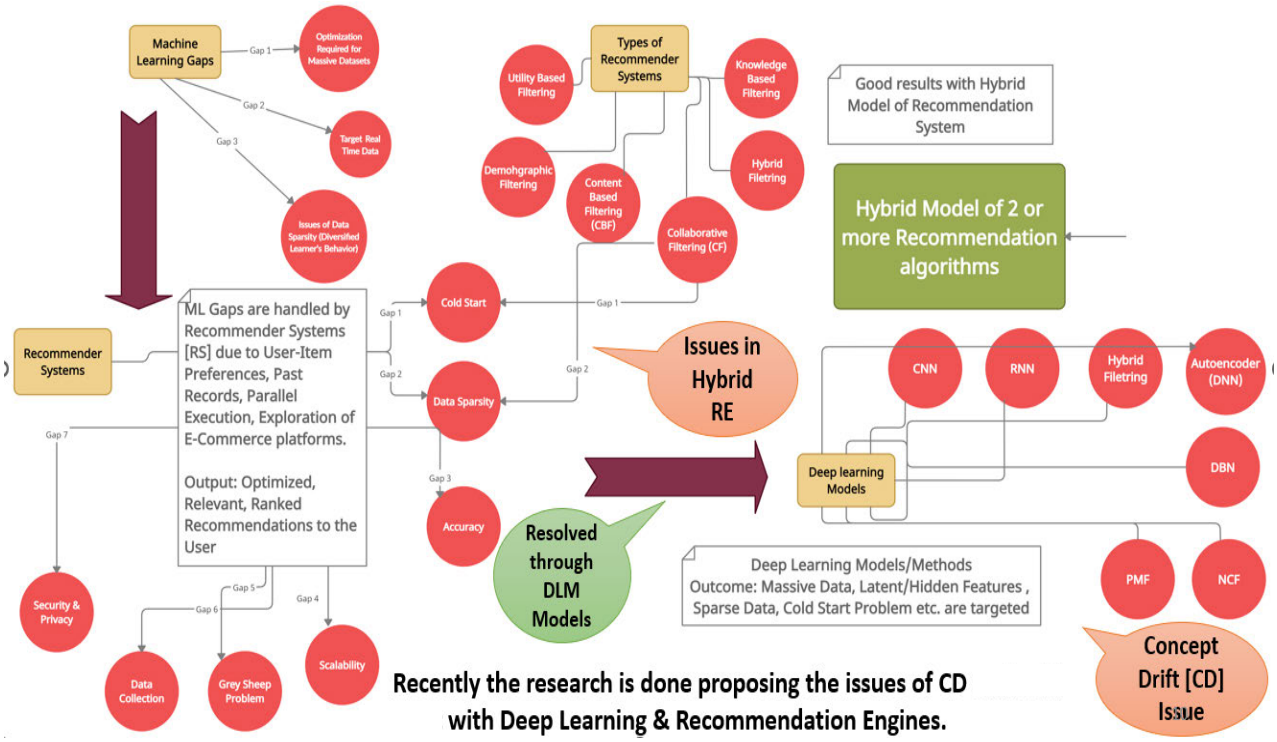


FIGURE 10. Gap analysis of systematic literature review.

TABLE 7 represents the literature review of hybrid DLMs, highlighting the issue of concept drift.

Concept Diagram of Hybrid Deep Learning Models handling the Concept Drift Issue:

The diagrammatic representation of the literature review on handling concept drift issues in deep learning models is shown in FIGURES 11 and 12.

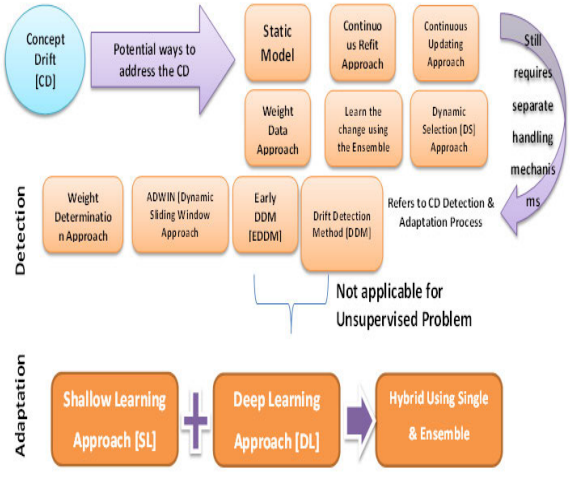


FIGURE 11. Concept diagram of handling concept drift issue.

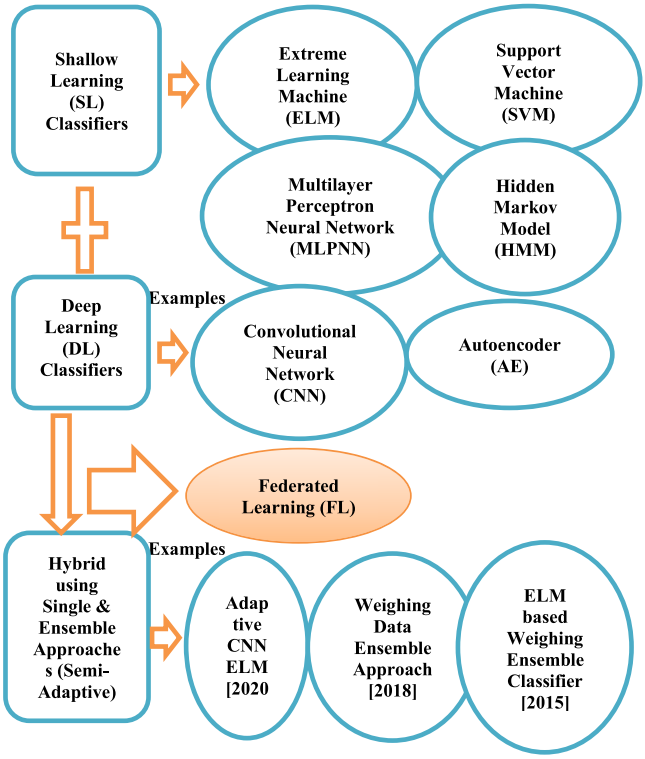


FIGURE 12. Concept diagram of adaptation of concept drift issue.

Important highlights of FIGURE 12 are as follows:

i) **Shallow Learning (SL) Classifiers** are simple, fast-processing, and can efficiently handle classification and regression problems in structured data only.

ii) **Deep Learning (DL) Classifiers** excel with complex and unstructured data, such as big data, and provide greater accuracy than SL classifiers due to their feature extraction mechanisms.

TABLE 7. Literature review of hybrid deep learning models and concept drift.

Authors	Year	Contribution
Syed Muslim Jameel, et.al [22]	2020	The review focuses on the taxonomy of concept drift, its adverse effects, and existing approaches to mitigate it as part of the systematic critical literature review. It was noted that the adaptive convolutional ELM (ACNNELM) algorithm, which falls under the hybrid deep learning (semi-adaptive) category, provides more accurate results [16,22].
Simone Disabato, et.al [23]	2021	The research focused on the Tiny Machine Learning approach, a recent field aimed at developing machine and deep learning mechanisms for use in embedded systems and Internet of Things (IoT) devices [23]. To address concept drift affecting the data-generating process, the Tiny Machine Learning for concept drift (TML-CD) method was developed. It combines a cross-adaptation module with deep learning feature extractors and a k-nearest neighbours' classifier [23].
Cuie Yang, et.al [24,33]	2021,2015	The study focused on concept drift in target learning (CDTL), which primarily involves adapting the target model and facts from source areas to a changing environment—a topic that has not been extensively explored in the literature [24,33]. A hybrid ensemble strategy was also proposed to address the CDTL problem, where data in the target area are generated using a flowing, chunk-by-chunk method from dynamic scenarios [33].
Manzoor Hashmani, et.al [25]	2020	The research discusses the issue of concept/data drift and its application areas in addressing current computing challenges. The three primary types of concept drift highlighted are virtual concept drift (VCD), real concept drift (RCD), and hybrid concept drift (HCD). managing RCD is more challenging than managing VCD, while HCD management remains an unresolved problem [25].
Chun-Cheng Lin, et.al [26]	2019	The study described an ensemble learning algorithm [15] designed to help offline classifiers manage three stages of condition-based maintenance (CBM), along with concept drifts and imbalanced data. Stage 1 (training an ensemble classifier) and Stage 3 (creating a new ensemble) utilise an improved Dynamic AdaBoost [15,26]. Stage 2 (detection of concept drifts in imbalanced data) employs an upgraded Linear Four Rates (LFR) technique, while the NC classifier and SMOTE address the imbalanced data. Experimental results on datasets with varying degrees of imbalance demonstrated that the proposed method effectively detects all concept drifts and achieves a high accuracy rate in detecting minority-class data, exceeding 94% [15,26].
Scott Wares, et.al [27]	2019	The study provided a comprehensive depiction of concept drift and its inherent data problems in stream mining. It also offered future research recommendations for concept drift detection and stream mining algorithms [27].

iii) **Federated Learning (FL)** offers a federated deep learning framework that ensures user data privacy, manages concept drift, and enhances communication efficiency.

Research Gaps in the related works:

Various tools and technologies are needed to tackle the challenges posed by big data [71]. Cloud storage and cloud computing offer promising solutions but are still in the early stages of development. Moving algorithms closer to the data is essential to avoid data transformation, which requires significant internet bandwidth.

Effective data management is critical, particularly for well-characterised archival data and heterogeneous datasets. Emerging database technologies require careful consideration when selecting the appropriate type.

The rise of big data has also driven the development of efficient data mining algorithms. Speed and efficiency are crucial for data exploration, and algorithms need to be parallelised, distributed, or optimised for GPU-based or cluster-based processing.

Collaboration among astronomers, statisticians, mathematicians, computer scientists, information scientists, and data scientists is essential to extract meaningful insights from massive datasets. Interdisciplinary and multidisciplinary collaboration is key to addressing complex challenges and maximizing the potential of big data in astronomy [71].

These studies [71] have extensively explored various big data frameworks for healthcare and transportation applications, implementing big data analytics and cloud computing services for large-scale applications.

IV. FEDERATED LEARNING

- The integration of on-device Artificial Intelligence, blockchain, and edge computing/IoT creates the mechanism for FL, where a centralised ML model is trained on distributed data [37].
- FL, as depicted in FIGURE 13, employs a method that encrypts user-sensitive information using a key not controlled by the central cloud server [37]. This approach, known as the Secure Aggregation Principle, allows the centralised server to protect and combine the encrypted

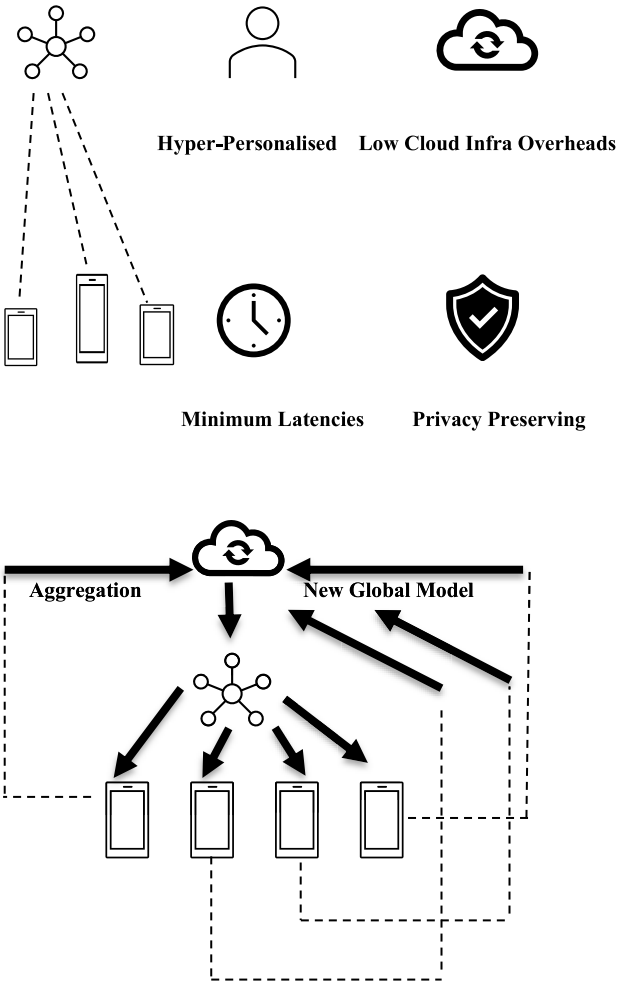


FIGURE 13. Overview of federated learning [37].

or encoded outcomes/results while being able to decode only the aggregated results. This function-based encryption is referred to as a zero-sum masking protocol [37].

- The zero-sum mechanism masks the sum to 1 in one direction and 0 in another. One direction handles the mixing and protection of the encoded user data, while the other decodes the training results/outcomes for the server. This process continues until completion, at which point the masks cancel each other out [37].

A. FEDERATED RECOMMENDER SYSTEMS [FEDRECSYS]

- Federated recommender systems (FedRecSys), as shown in FIGURE 14 [41], are a federated learning component focused on decentralised suggestions. Compared to traditional recommender systems (RecSys), FedRecSys [35] significantly enhances user privacy and data security by decentralizing private user data at each party [35].
- Each party or organization can function as a data provider or a recommender system (RecSys). The three main components of a RecSys entity are user profiles, item attributes, and rating information. A data provider entity typically contains additional user profiles or item attributes [35].

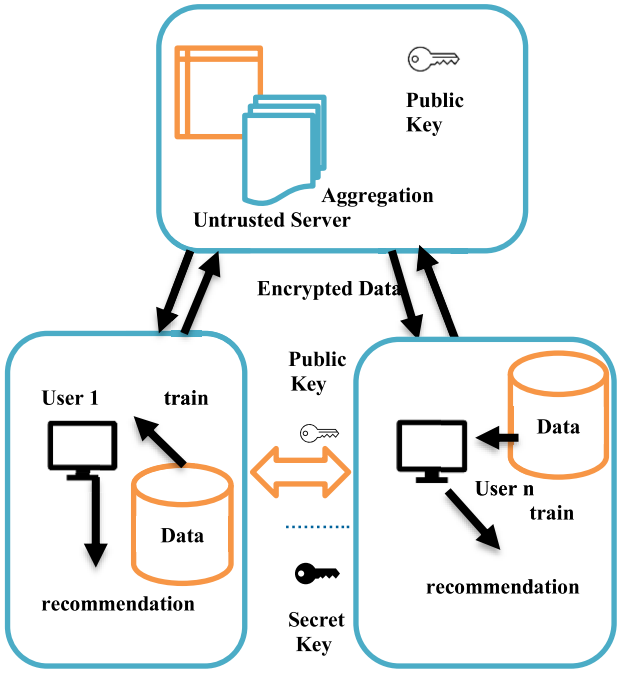


FIGURE 14. Federated recommender system [41].

FedRecSys can be categorised into Horizontal FedRecSys, Vertical FedRecSys, and Transfer FedRecSys [35], based on the sharing state of users and items, as shown in FIGURE 15 [67].

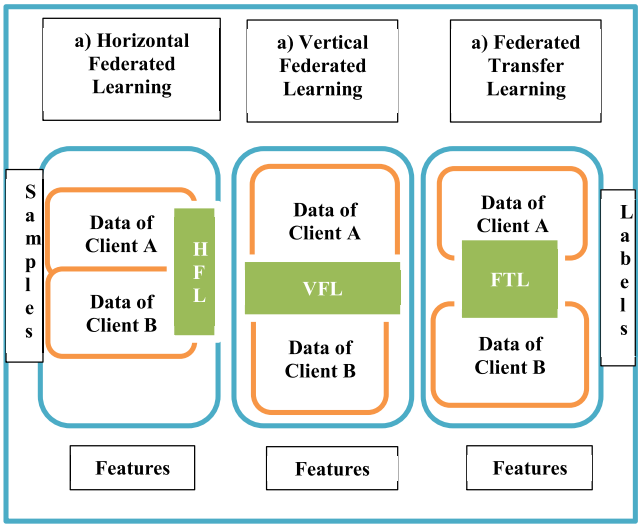


FIGURE 15. Types of federated learning [67].

- The system is known as the Horizontal FedRec System when items are shared but users vary across parties. In this scenario, the parties could be individual users or groups of people [35].

- The Vertical FedRec System is used when two parties share the same user set but have separate item sets or feature spaces. Here, the parties could be different data providers or recommenders [35].
- The Transfer FedRec System is introduced when users or items are not shared between parties. Typically, the parties are distinct recommender systems [35].

B. LITERATURE REVIEW: FEDERATED RECOMMENDER SYSTEMS

TABLE 8 presents a detailed literature review of FedRecSys.

Challenges/research Gaps in Implementation of Hybrid Federated Deep Learning Models:

The challenges and research gaps in the development of the Hybrid Federated DLM as explained in the manuscript are as follows:

1) Adaptation of the main Federated DLM, focusing on client devices' updated item profiles (encrypted item gradients) and aggregating user item gradients at the decentralised server side for the next iteration of improved recommendations.
2) Provision of security for the recommender system framework by maintaining the privacy of user's data.
3) Ensuring the accuracy and relevance of recommendations to the end user through performance measures such as mean absolute error (MAE) and root mean square error (RMSE).

Researchers and practitioners can address these challenges through the development of the novel FedRecSys framework (hybrid federated DLM for recommender systems), which integrates the Federated Learning approach with a CRE to address the critical issue of concept drift in recommendation systems.

The hybrid recommender system, or federated DLM, pertains to the Hybrid framework of the FedRecSys, where FL (a new dimension of DL) is applied to the collaborative recommendation system. In the literature review, the CRE is shown to be the most effective and preferred RE compared to others, as it operates on 3-dimensional data:

user data, items, and ratings (clicks, reviews, buying/purchasing patterns/behaviour).

Thus, this hybrid approach not only addresses the major concept drift issue in REs but also provides accurate and relevant recommendations to the end user, resulting in improved recommendation performance.

V. RESEARCH METHODOLOGY

FIGURE 16 illustrates the detailed flow of the project, encompassing data collection, design, analysis, implementation, and results comparison.

A. DESCRIPTION

The system will receive a large dataset, such as the Amazon ratings/reviews dataset collected from online shopping of various products/items. The collaborative filtering recommendation system will be applied to Amazon's extensive collection of user evaluations. This system employs matrix factorization techniques, as illustrated in FIGURE 17. Collaborative filtering algorithms used in recommendation systems leverage this matrix factorization technique.

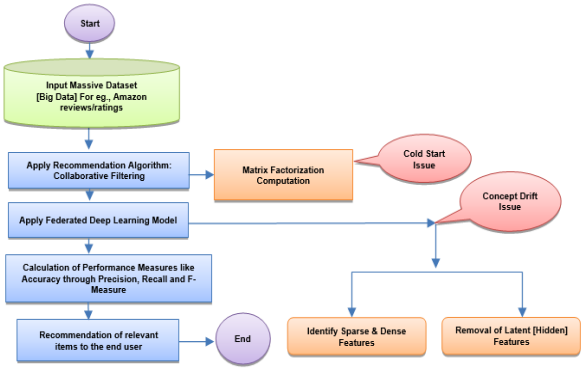


FIGURE 16. Flowchart of research methodology.

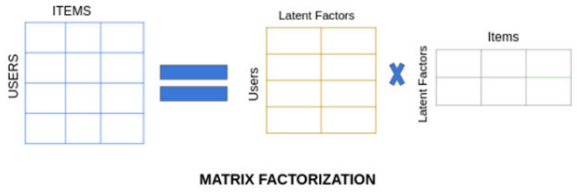


FIGURE 17. Matrix factorization structure.

CNNs and AEs, for example, benefit from identifying both sparse features (mostly zero values and categorical data) and dense features (non-zero values and continuous information) and from removing latent (hidden) features. However, DLMs face the challenge of concept drift, which involves variations or changes in data distribution over time.

B. CONCEPT DRIFT

Concept drift can either introduce a completely new idea or involve an existing one. Drift recurrence refers to when a drift that was previously present reoccurs [1]. Since understanding the prior concept is sufficient, the drift recurrence mechanism is more complex in adapting to the new idea [1]. A common example of drift recurrence is the customer decision to purchase clothing; the desire to buy clothes returns or reemerges each winter [1], [22].

Types of Concept Drift: (with the predictive point of view):

Different types of concept drift for the prediction are presented in FIGURE 18 [22].

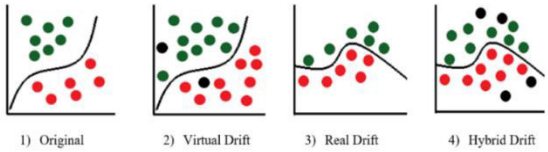


FIGURE 18. Types of Concept Drift (Predictive point of view).

TABLE 8. Literature review of federated recommender systems.

Authors	Year	Contribution
Guanyu Lin, et.al [31,34]	2022	In this article, FedRecSys is described and examined in the context of a fundamental matrix factorization model, such as PMF. It was noted that more sophisticated factorization-based models, like SVD++, can be integrated with the proposed framework and learning methods [34]. For the experimentation, two open MovieLens datasets from GroupLens (https://grouplens.org/) were used. The first dataset, MovieLens 100K, includes 100,000 user ratings for 1,682 films from 943 users. The second dataset, MovieLens 1M, comprises 100,0209 user ratings for 3,952 films [34]. The evaluation metrics for performance included mean absolute error (MAE) and root mean square error (RMSE), which are widely used in the recommender systems community [34].
H. Chen, et.al [35]	2022	In this study propose an efficient and secure recommendation system based on a federated matrix factorization mechanism. In this approach, each user locally trains the model using their own data and uploads the resulting gradient vector instead of the original data [35]. A central server performs model updates and verifies the aggregation results sent by other servers, all while maintaining user privacy [35]. Participants' behaviour was monitored to determine incentives or penalties, and users are allowed to withdraw from the training process at any time [35].
Z. Jie, et.al [10,36]	2022	In this work, a recommendation system model based on user embedding features is proposed [36]. According to the research, the server can utilise user embedding features for tailored recommendations while maintaining user data privacy. Clients can use their original data for local personalised recommendations [36]. The MovieLens-1M dataset was employed for the studies. Experimental results demonstrated that the proposed federated learning strategy outperforms the established federated learning approach [36].
Vasileios Perifanis, et.al [37]	2022	In this study, a decomposition of the aggregation step into matrix factorization and neural network-based averaging mechanisms was proposed to enhance the method [37]. It was demonstrated that FedNCF achieves comparable recommendation quality to the original NCF system through experimental validation, while the proposed aggregation results in faster convergence compared to current techniques [37]. The FedNCF [31] architecture includes three federated learning algorithms: a generalised matrix factorisation (FedGMF), a multilayer perceptron (FedMLP), and a combination of FedGMF and FedMLP (FedNeuMF). These models represent federated adaptations of the related NCF approach. The datasets used for experimentation were MovieLens 100k, 1M, LastFM, and Foursquare [31,37].
Waqar Ali, et.al [38]	2021	A federated learning-based, privacy-preserving CF model for context-aware recommender systems, known as FedCARS, is described in this research study. Instead of aggregating users' personal information on a central server, the data are divided into two separate components: user data and sharable item information [38]. The performance of the proposed algorithm was evaluated using two publicly available datasets, Ml 100K and Ml-latest-small, with both prediction and ranking approaches [38]. FedCARS was compared with Bayesian personalised ranking (BPR), most popular recommender (POP), User-KNN, and Item-KNN [38]. Performance metrics such as MAE and RMSE [38] were used for evaluation.
Pan Zhou, et.al [39]	2021	A privacy-preserving dynamic recommendation system (DRS) that models each service provider as a distributed online learner with context awareness is discussed in this research paper [39]. The study also proposes the use of the tree-based privacy preserving distributed online learning method (T-PriDO) [39]. The experimentation utilised the YFCC100M dataset, which was supplied by Yahoo in 2014. This dataset comprises over 100 million media items, including 100.2 million photos and 0.9 million movies, with each entry containing basic information such as location, owner name, camera, title, media source, and tags [39].
Enyue Yang, et.al [40]	2021	In this study, a heterogeneous collaborative filtering (HCF) approach was applied, utilizing various types of user feedback, such as 5-star numerical ratings and binary like-or-dislike evaluations, in a privacy-aware manner [40]. Additionally, a new and universal federated matrix factorization algorithm, called federated collective matrix factorization (FCMF), was developed for HCF [40]. The primary objective of FCMF was to leverage diverse feedback data to accurately approximate user preferences while ensuring privacy protection. Users' latent feature vectors and original rating data are kept locally, with latent feature vectors for low-sensitivity items serving as a bridge for joint training [40]. Homomorphic encryption (HE) and differential privacy (DP) techniques were employed to secure the privacy of participants during collaborative training [40]. The datasets used include MovieLens 100K, MovieLens 1M, MovieLens, and Netflix.
Yongjie Du a, et al [32,41]	2021	Rather than using raw data, a user-level distributed matrix factorization framework is presented, where the model is learned by collecting gradient information from users [32]. The datasets used are Filmtrust and Movielens-100K. Security is ensured through Homomorphic Encryption.
Bichen Shi, et al [19]	2021	A technique known as DARES—the distributed recommender system algorithm—is introduced in this research paper. It is based on the asynchronous advantage actor-critic model (A3C) model and incorporates reinforcement learning [19]. The DARES algorithm integrates A3C and federated learning (FL) techniques, allowing users to keep their data locally on their own devices. The system architecture consists of a global recommendation model, trained on a central server using model updates computed on user devices, and a local recommendation model, trained locally on the user devices using their interaction data [19].
Liu Yang, et al [43]	2020	The empirical evidence of the Concept Drift issue was summarised and analysed in this research to assess its applicability to recent computing advances [43]. Three basic types of CD were examined: hybrid concept drift (HCD), real concept drift (RCD), and virtual concept drift (VCD). Managing RCD is more challenging than managing VCD, while HCD management remains an undetermined problem [43].

TABLE 8. *(Continued.)* **Literature review of federated recommender systems.**

Chaochao Chen, et al [44]	2020	A new privacy-protecting POI recommendation system, PriRec, was created. First, users' private information (features and actions) is kept on their own devices, such as cell phones or tablets, to ensure data privacy. Second, PriRec utilises Factorization Machine representations, which integrate a linear model with a feature interaction model. Security mechanisms employed include DP techniques, HE, and multi-party computation (MPC).
Ben Tan, et al [45]	2020	Developed a federated recommender system for content recommendation, which utilises federated learning techniques.
Amir Jalalirad, [46]	2019	An effective FL extension for recommender systems, which enhances personalization and explores closely related meta-learning techniques, was introduced [46]. The outcomes were compared with those of a federated meta-learning neural network. The authors presented two iterations of their algorithm—NN-SELF and NN-MIXED—and reported RMSE values for various quantities of support data using the MovieLens100k dataset [46]. The RMSE values for the algorithms were as follows: SVD++(SELF): 1.05, SVD++(MIXED): 1.07, NN(SELF): 1.35, NN(MIXED): 0.98, Proposed Method: 1.03.
Di Chai, et al [47]	2019	The first FL-based secure matrix factorization framework, known as FedMF, was introduced and designed to address the limitations of conventional obfuscation or encryption-based techniques [47]. FedMF was implemented using a user-level distributed matrix factorization framework, where each user uploads only gradient information (rather than raw preference data) to the server for model learning [47]. Homomorphic encryption was integrated into the distributed matrix factorization framework to enhance security [47]. The MovieLens dataset, which includes 100K ratings submitted by 610 individuals for 9724 films, was used for the experimentation [47].
Lei Zhou, et al [48]	2012	A federated recommender system was presented that generates personalised recommendations using data from multiple online learning platforms. The study considers the online learning and collaboration platform 'Graasp' and builds upon the 3A Interaction Model—Actors, Assets or Resources, and Activity Spaces. The FedRecSys architecture includes a harvester, an RE, and a widget.

Eq. 1 defines concept drift between time point t_0 and time point t_1 [35], as given below:

$$\exists \mathbf{X} : \mathbf{pt_0}(\mathbf{X}, \mathbf{y}) \neq \mathbf{pt_1}(\mathbf{X}, \mathbf{y}) \quad [35] \quad (1)$$

Here, pt_0 denotes the joint distribution of the input variable set X and target variable y, set at time t_0. Concept drift incidence can be described using three methods [30], [35].

- prior probabilities of classes, p(y) may change over time.
- class-conditional probability distributions, p (X, y) might change
- posterior probabilities p(y|X) might change.

C. REAL DRIFT

It highlights the fluctuation in the prior probability of a class. For example, if a student frequently reads sports news, it indicates an interest in sports. However, if the student's interest shifts to political news, perhaps while writing a report, this change—despite the consistent distribution of the two topics—is known as 'real drift' [30].

D. VIRTUAL DRIFT

Virtual drift refers to changes in the distribution of incoming data. For instance, if a user is interested in sports news, the focus of sports news may shift from topics like football, basketball, and handball to the current major sporting event. Consequently, although the distribution of sports news changes, the user's interest remains unchanged [22]. Other types of concept drift include Sudden, Incremental, Continuous, Gradual, Reoccurring, and Hybrid Drifts, which relate to the prediction of time.

Types of Concept Drift: (from the point of view of time):

Different types of concept drift with respect to time are presented in FIGURE 19.

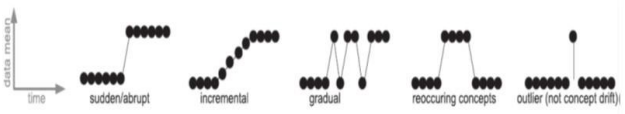


FIGURE 19. Types of Concept Drift (from the point of view of time).

- **Sudden Drift:** Occurs when one concept (C1) is abruptly replaced by another concept (C2). For example, a researcher focused on big data storage and retrieval algorithms might suddenly be tasked with preparing a survey on security issues in big data. This abrupt shift in focus from storage and retrieval to security issues represents a sudden drift.
- **Gradual Drift:** Occurs when a new concept (C2) begins to grow in prominence while the original concept (C1) is still of interest. Over time, C2 becomes the dominant concept while C1 gradually declines until it eventually disappears. For instance, a researcher initially interested in big data storage and retrieval algorithms might gradually develop an interest in security issues in big data. Eventually, the researcher becomes fully focused on security issues and stops studying storage and retrieval algorithms.
- **Incremental Drift:** Can only be identified over an extended period as small, gradual changes accumulate. For example, if members of the big data community start focusing more on security issues rather than

storage and retrieval algorithms, over time, security issues will capture their primary attention while storage and retrieval algorithms fall out of focus. Incremental drift typically involves one active concept at a time, whereas gradual drift involves two concurrently active concepts.

- **Re-occurring Drift:** Refers to the reappearance of a previously dominant concept after a period of absence. For example, a researcher might shift attention from big data storage and retrieval algorithms to security issues. If the researcher later finds security issues less relevant, they might return to focusing on storage and retrieval algorithms.

In the context of federated learning, DLMs hold significant promise for capturing and adapting to various types of concept drift. By leveraging a combination of specialised algorithms, communication strategies, and adaptive learning techniques, DL models in federated settings show considerable potential in identifying and addressing different forms of concept drift. Despite challenges such as communication overhead and heterogeneous data, concept drift can be effectively managed in decentralised systems. Continuous advancements in federated learning algorithms and privacy-preserving methods are facilitating these improvements.

Concept Drift in REs:

- Regarding REs, as additional users, products, and rating information are consistently added to the system, various changes occur in the critical interactions between users and the recommended items. This issue, known as concept drift or temporal dynamics, reflects the evolving nature of REs [4], [31].
- Dynamic recommendation systems (DRSs) are designed to adapt their recommendations to meet user needs in response to time-based changes in various domains, such as user or item-related data and other time-sensitive events [4], [31].
- DRSs, like traditional static recommendation systems, employ common filtering algorithms: content-based (CB), collaborative filtering (CF), and hybrid mechanisms [4].
- The application of time-based deep learning mechanisms to address time-based dynamics in recommendation systems warrants further exploration. Recent advancements in DLMs have led to various approaches for handling concept drift [4], [31].

1) Recurrent neural networks (RNNs) are used to manage concept drift for time series anomaly detection, addressing issues arising from sudden or recurring changes in typical behaviour. As new data becomes available, the model is incrementally trained to adjust to variations in data distribution [4], [31].
2) Evolutionary neural network models are employed for time-evolving text classification. In the field of recommendation systems, significant efforts have been made to enhance system performance using deep learning approaches [4], [31].

Adaptation and Ensemble Learning:

- The concept of 'adaptability' was developed to mitigate performance loss caused by concept drift in ML models. Adaptability refers to the ability of machine learning models to dynamically update themselves in response to changes in the data.
- The 'Adaptation' method allows ML models to be tuned or self-regulate to adjust to new ideas. This strategy helps to mitigate performance degradation due to its dynamic nature [16].
- To address issues of concept drift and unbalanced data, ensemble learning [15] combines multiple classifiers to provide a high-performance solution [15], [25].
- Ensemble learning [17] is a supervised ML technique that involves a 'committee of experts' (i.e. multiple ML models) to make decisions based on a specific voting system [17].
- Unlike deterministic forecasting techniques, which rely on a single model, ensemble learning combines predictions from multiple models into a single forecast outcome [17], [25].
- Studies have shown that ensemble learning often outperforms any single forecasting model, improving classification performance in the presence of concept drift [1], [17].
- Ensemble learning is applicable in both online (non-stationary) and offline machine learning environments and is suitable for problems involving large amounts of complex data [17].
- In an ensemble approach, the final decision is made by aggregating the outputs of various classifiers. Techniques such as Max Voting, Averaging, Stacking, Blending, Bagging, and Boosting are used to combine the decisions of each model [1], [25].

VI. LITERATURE REVIEW/SURVEY OF RECOMMENDATION SYSTEMS DATASETS

TABLE 9 presents the detailed literature review of various well-known datasets required for REs:

VII. DATASET DESIGN

Amazon Reviews/rating Dataset [28]:

About the Dataset

The Datafiniti Product Database includes over 34,000 customer reviews for various Amazon products, such as the Kindle, Fire TV Stick, and others. The dataset contains basic product details, ratings, review content, and additional information [23], [28].

Use Case of the Data

- Analyse Amazon's most popular consumer electronics product launches, gain insights into customer feedback, and support ML models.
- Determine the most reviewed Amazon products [23], [28].
- Determine the number of initial and ongoing customer reviews for each product [23], [28].

TABLE 9. Literature review of various datasets for RE.

Authors	Dataset Details	Description	Purpose	Results
R. Manikandan, et al. [14]	**Chronic obstructive pulmonary disease (COPD)** health records were collected from the website of a public hospital. The data were categorised by geographical regions: Northeast, North Central, West, and South. The dataset also includes multimedia files such as audio, video, and images. Key components of the healthcare records with multimedia data include personal health information, self-monitoring health values, educational resources, and coaching for achieving personal health-related goals. The primary focus is on the first six weeks of log data following the release, which consists of a total of 160 log files.	The datasets collected account for all-cause and COPD-related hospital readmission rates within 30 days following the index date. The demographic factors included in this group are age, sex, geographic location, rural versus urban context, and type of health insurance (Medicare or not).	The results demonstrated that the proposed PASO-WPR system performed well in terms of accuracy, including overall accuracy measures. When compared to previous item-based collaborative filtering recommendation systems, the PASO-WPR system's coverage was found to provide more valuable information.	Cleaned sessions from each dataset were separated, with approximately 25% randomly selected as the test set and the remaining 75% used as the training set. The effects of various recommender systems were then investigated. Specifically, the study developed and evaluated three different recommender methods: Weighted Association Rule Mining (WARM), PreOrder Linked Web Access Pattern (PLWAP) Tree Mining, Social Recommender System (SRS), and the proposed PASO-WPR system.
Abhaya Kumar Sahoo, et al. [15]	**Movie Lense dataset**	The dataset was used to create an item-based collaborative filtering recommendation system.	A hybrid item-based collaborative filtering model was developed by optimizing the neighbour parameters [20]. In this context, 'Precision' refers to the percentage of recommended products or items that have been purchased, while 'Recall' represents the percentage of products or items that were recommended [20].	To assess the performance of the recommender system, several metrics were evaluated. The rating matrix was converted into binary form after normalizing the relevant data. Results were interpreted with the understanding that accuracy improves as RMSE decreases [20]. This approach enhances the quality of recommendations. Performance metrics such as recall and precision were employed to evaluate the model's accuracy [20].
Christos Sardianos, et al. [16]	A portion of the **Epinions dataset** was used for the study. Epinions [9] is a well-known product review site, and the dataset, provided by the University of Koblenz-Landau, includes 13,668,320 ratings.	**Bipartite Graph Characteristics of the Epinions dataset:** 120,492 distinct users (raters) Number of Unique Items: 755,760 Avg. outDegree/User 113.44 Avg. inDegree/User: 13,668,320 Sparsity: 1.50 ×104	A method was introduced to increase the number of edges in graph partitions to manage their sparsity (ratings) [9]. The experimental evaluation aimed to assess the effectiveness and development of alternative approaches for selecting edges to replicate across partitions. Additionally, it compared the proposed methods with a standard random partitioning approach without overlaps and with a state-of-the-art	The performance of three proposed strategies was evaluated: • Active Users All Ratings (AUAR) [9]: Replicates all of a user's ratings across all partitions, leading to the addition of new items to the partitions indirectly. • Active Users Selected Ratings (AUSR) [9]: Copies only selected ratings from chosen users, adding additional ratings to a partition's items without introducing new items. • Informative Users Selected Ratings (IUSR) [9]: Replicates selected ratings from users

TABLE 9. *(Continued.)* **Literature review of various datasets for RE.**

			collaborative filtering algorithm (SVD++) applied to the entire, unpartitioned bipartite graph (rating matrix) [9].	chosen to enhance ratings for less-rated items. The results of the baseline partitioning approach, labelled 'Random' in the plots, were also discussed. This approach randomly divides the bipartite graph (i.e., the set of rating edges) into a predetermined number of partitions [9].
Raphael Louca, et al. [17]	**Etsy.com**	The Etsy.com dataset is a training set comprising implicit feedback data collected over one day from an item-to-item recommendation module installed on Etsy.com item pages. The feature set includes item features (e.g., item purchase count) as well as cross features (e.g., tf-idf similarity between target and candidate items) [3].	In the suggested model, high-priced items are ranked higher in the relevance hierarchy compared to low-cost items.	The proposed revenue model [3] outperforms all other models in terms of AUC and Profit@k, where Profit@k represents the profit generated by the top k highest-ranked items [3]. **Area Under the Curve (AUC) [3]:** AUC measures the area under the receiver operating characteristic curve and evaluates relevance. It indicates the likelihood that a randomly selected positive instance will be ranked higher than a negative instance by the classifier [3].
Deuk Hee Park, et al. [18]	**Large massive datasets: Movie dataset and Amazon**	Researchers extensively use movie datasets to develop hybrid approaches in recommendation systems. In this context, the proposed method demonstrates superior accuracy compared to the standard Slope One algorithm, thanks to the use of the Amazon dataset [10].	A thorough analysis of five conventional algorithms for recommender systems in electronic commerce is proposed, highlighting the benefits and drawbacks of each ranking method in the chosen paper [10]. The paper also discusses the challenges associated with recommender systems in e-commerce [10].	The five fundamental methods—CBF, CF, DBF, KBF, and hybrid—are discussed, with each model's benefits and drawbacks illustrated [10]. Additionally, the paper provides a comparison of the chosen methods concerning various issues, including scalability, response time, data sparsity, accuracy, performance, and security [10].
Neha Verma, et al. [19]	**Survey of E-Commerce Portals**	The paper describes the role of data mining in the recommendation system along with the RS workflow [2].	The paper compares recommendation systems of several e-commerce websites, evaluating RS methodologies and addressing related problems [2].	The top-rated websites are compared with respect to: a) recommendation system or engine techniques b) products and services recommended by the websites [2].

- Compare reviews published within the first 90 days of a product introduction and the item's cost [23], [28].
- Evaluations in the first 90 days following a product launch compared to the duration the product has been on sale [23], [28].
- Map review text keywords to reviewer ratings to enable sentiment model training [23], [28].

Applications of Hybrid Federated Deep Learning Models

Hybrid federated DLMs offer versatile applications across various domains for personalised recommendations while ensuring privacy and scalability. Key research domains where Hybrid Federated DLM finds real-world applications include Federated Recommendations and Federated Computer Vision. The real-world applications of Hybrid Federated DLM are listed below:

- Movie Recommender Systems
- News Recommender Systems
- Smart City Surveillance
- Mobile Devices through Natural Language Processing
- E-Commerce and Social Media through Matrix Factorization (Singular Value Decomposition)
- Financial Applications
- Healthcare

Hybrid federated DLMs have diverse applications in personalised recommendations across multiple domains such as e-commerce, healthcare, news and media, financial services, education, and e-learning. These models harness collective insights from distributed data sources while maintaining user privacy. By doing so, they facilitate personalised experiences, improve decision-making processes, and foster innovation across various industries.

Scope of the Study

The proposed system will focus solely on recommending items to users by applying a selected DLM within REs. It addresses key challenges in REs, such as data sparsity, and in DLMs, including concept drift detection, which can be integrated into the DL RE. The novelty of this system lies in the hybridization of Federated DL with REs, leading to the development of the 'Federated Recommender System (FedRecSys)'.

The performance of the system will be evaluated using metrics such as accuracy, RMSE, and MAE, which aim to enhance user satisfaction and trust.

Challenges of the Proposed Study: The challenges in the design and implementation of Hybrid Federated DLM for accurate recommendations are as follows:

1) Selection of the concept drift-detection method with a comparative analysis.
2) Analysis of the adaptation process for concept drift, considering different types of FL (Vertical, Horizontal, and Transfer).
3) Comparative analysis of security mechanisms in federated environments, including Homomorphic Encryption, Differential Privacy, and Secure Multi-Party Computation.
4) Result analysis using various performance measures and benchmark datasets.

Researchers and practitioners can overcome these challenges with the development of this novel proposed framework of FedRecSys (Hybrid Federated DLM for Recommender Systems), which will incorporate the Federated learning approach and collaborative recommendation engine to handle the most important Concept drift issue in recommendation system.

VIII. CONCLUSION

The study reveals that significant research opportunities exist in Recommendation Engines (REs) and Federated Deep Learning Models (FDLMs). Recommendations are generated through various e-commerce platforms such as MovieLens, Epinions, Amazon Ratings, and others. Different types of REs are examined in detail.

A. The proposed work focuses on a systematic literature review of Machine Learning and Information Retrieval, including RE, RE Datasets, RE Challenges, Collaborative RE, Deep Learning Models and RE, and Federated Recommender Systems. It addresses and manages various issues, particularly Concept Drift, through the hybridization of RE and FDLMs.
B. The proposed work outlines the advantages and limitations of various comparative studies, highlighting that Collaborative RE generally outperforms other types of RE.
C. The comparative study of top-rated websites and e-commerce portals examines which DLMs are employed by different e-commerce and social media platforms.
D. The literature review of RE and Collaborative RE discusses the various issues handled by REs, emphasizing that Concept Drift is the most significant issue requiring detailed attention.
E. The systematic literature review of DLMs and Federated Learning for managing Concept Drift introduces a novel hybrid approach for detecting and adapting to Concept Drift, which is an emerging area of research in REs.
F. The proposed work will be evaluated using performance measures such as Accuracy, RMSE, and MAE, which aim to enhance user satisfaction and trust.

FUTURE RESEARCH DIRECTIONS

Future Research Directions of the research work can be as follows:

1. Detection and adaptation of various issues in REs using multiple supervised methods.
2. Provision of security to the recommender system framework by maintaining user data privacy.
3. Ensuring the accuracy and relevance of recommendations to the end user through performance measures such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and other relevant metrics.

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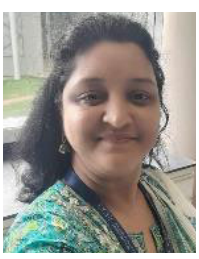
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